SAVEETHA SCHOOL OF ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

INSTITUTE OF PLACEMENT AND TRAINING

CSA09 –JAVA PROGRAMMING

1. Write a program to count all the prime and composite numbers entered by the user.

   Sample Input:

   Enter the numbers

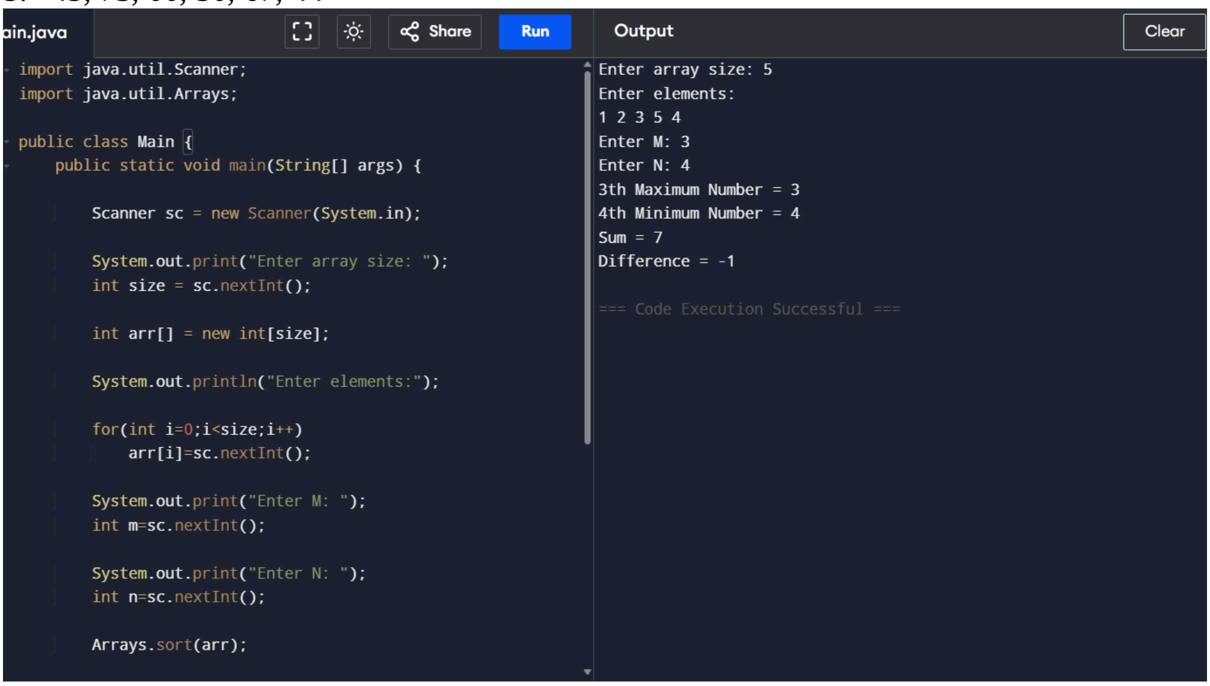
   4
   54
   29
   71
   7
   59
   98
   23

   Sample Output:

   Composite number:3
   Prime number:5

   Test cases:

   1. 33, 41, 52, 61,73,90
   2. TEN, FIFTY, SIXTY-ONE, SEVENTY-SEVEN, NINE
   3. 45, 87, 09, 5.0 ,2.3, 0.4
   4. -54, -76, -97, -23, -33, -98
   5. 45, 73, 00, 50, 67, 44



2. Find the M^{th} maximum number and N^{th} minimum number in an array and then find the sum of it and difference of it.

   Sample Input:

   Array of elements = {14, 16, 87, 36, 25, 89, 34}

   M = 1

   N = 3

   Sample Output:

1stMaximum Number = 89
3rdMinimum Number = 25
Sum = 114
Difference = 64

Test cases:

1. {16, 16, 16 16, 16}, M = 0, N = 1
2. {0, 0, 0, 0}, M = 1, N = 2
3. {-12, -78, -35, -42, -85}, M = 3 , N = 3
4. {15, 19, 34, 56, 12}, M = 6 , N = 3
5. {85, 45, 65, 75, 95}, M = 5 , N = 7

```java
import java.util.Scanner;

public class Main {
    public static void main(String args[]) {

        Scanner sc = new Scanner(System.in);

        int total = 0;

        for(int i=1;i<=4;i++)
        {
            System.out.print("Enter Denomination: ");
            int denomination = sc.nextInt();

            System.out.print("Enter Number of Notes: ");
            int notes = sc.nextInt();

            total = total + (denomination * notes);
        }

        System.out.println("Total Available Balance in ATM: "
            + total);
    }
```

```
Enter Denomination: 5
Enter Number of Notes: 3
Enter Denomination: 6
Enter Number of Notes: 2
Enter Denomination: 6
Enter Number of Notes: 8
Enter Denomination: 4
Enter Number of Notes: 8
Total Available Balance in ATM: 107

=== Code Execution Successful ===
```

3. Write a program to print the total amount available in the ATM machine with the conditions applied.
   Total denominations are 2000, 500, 200, 100, get the denomination priority from the user and the total number of notes from the user to display the total available balance to the user
   Sample Input:
   Enter the 1st Denomination: 500
   Enter the 1st Denomination number of notes: 4
   Enter the 2nd Denomination: 100
   Enter the 2nd Denomination number of notes: 20
   Enter the 3rd Denomination: 200
   Enter the 3rd Denomination number of notes: 32
   Enter the 4th Denomination: 2000
   Enter the 4th Denomination number of notes: 1
   Sample Output:
   Total Available Balance in ATM: 12400

   Test Cases:
   3 Hidden Test cases (Think Accordingly based on Denominations)

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```java
public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.println("1.String Palindrome");
        System.out.println("2.Number Palindrome");

        int choice = sc.nextInt();

        if(choice==1)
        {
            String str=sc.next();
            String rev="";

            for(int i=str.length()-1;i>=0;i--)
                rev=rev+str.charAt(i);

            if(str.equals(rev))
                System.out.println("Palindrome");
            else
                System.out.println("Not Palindrome");
        }
```

```
1.String Palindrome
2.Number Palindrome
```

4. Write a program using choice to check

    Case 1: Given string is palindrome or not

    Case 2: Given number is palindrome or not

    Sample Input:

    Case = 1

    String = MADAM

    Sample Output:

    Palindrome

    Test cases:

    1. MONEY
    2. 5678765
    3. MALAY12321ALAM
    4. MALAYALAM
    5. 1234.4321

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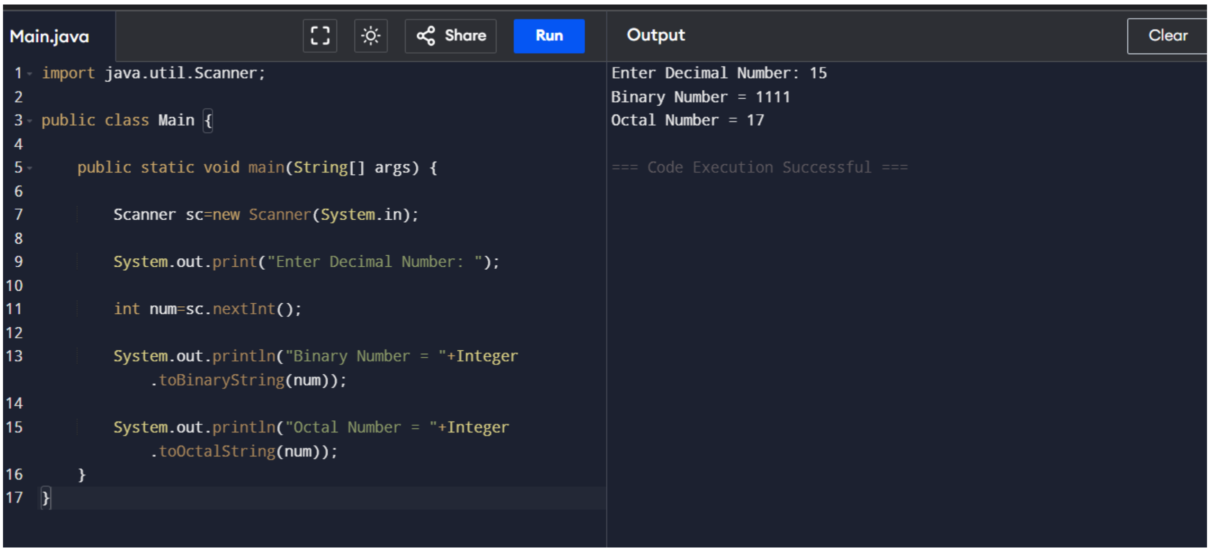
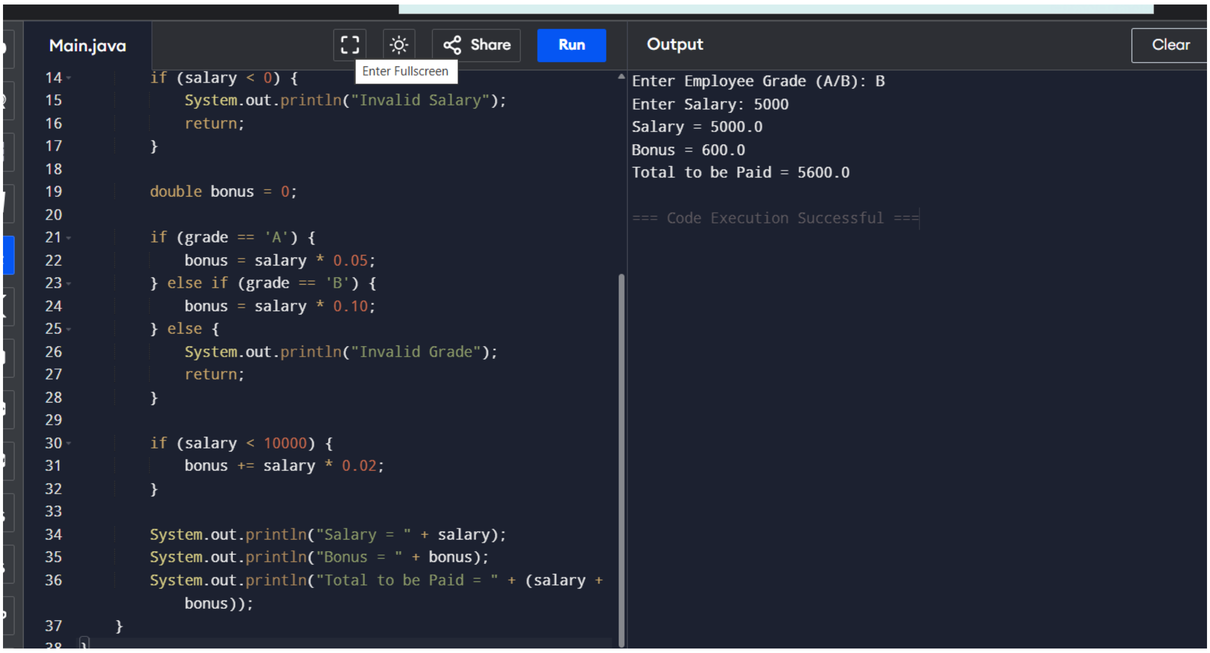



```java
import java.util.Scanner;

public class Main {

    public static void main(String[] args) {

        Scanner sc=new Scanner(System.in);

        System.out.print("Enter Decimal Number: ");

        int num=sc.nextInt();

        System.out.println("Binary Number = "+Integer
            .toBinaryString(num));

        System.out.println("Octal Number = "+Integer
            .toOctalString(num));
    }
}
```

```
Enter Decimal Number: 15
Binary Number = 1111
Octal Number = 17

=== Code Execution Successful ===
```

5. Write a program to convert Decimal number equivalent to Binary number and octal numbers?
   Sample Input:
   Decimal Number: 15
   Sample Output:
   Binary Number = 1111
   Octal = 17
   Test cases:

1. 111
2. 15.2
3. 0
4. B12
5. 1A.2




```java
        if (salary < 0) {
            System.out.println("Invalid Salary");
            return;
        }

        double bonus = 0;

        if (grade == 'A') {
            bonus = salary * 0.05;
        } else if (grade == 'B') {
            bonus = salary * 0.10;
        } else {
            System.out.println("Invalid Grade");
            return;
        }

        if (salary < 10000) {
            bonus += salary * 0.02;
        }

        System.out.println("Salary = " + salary);
        System.out.println("Bonus = " + bonus);
        System.out.println("Total to be Paid = " + (salary +
            bonus));
    }
}
```

```
Enter Employee Grade (A/B): B
Enter Salary: 5000
Salary = 5000.0
Bonus = 600.0
Total to be Paid = 5600.0

=== Code Execution Successful ===
```

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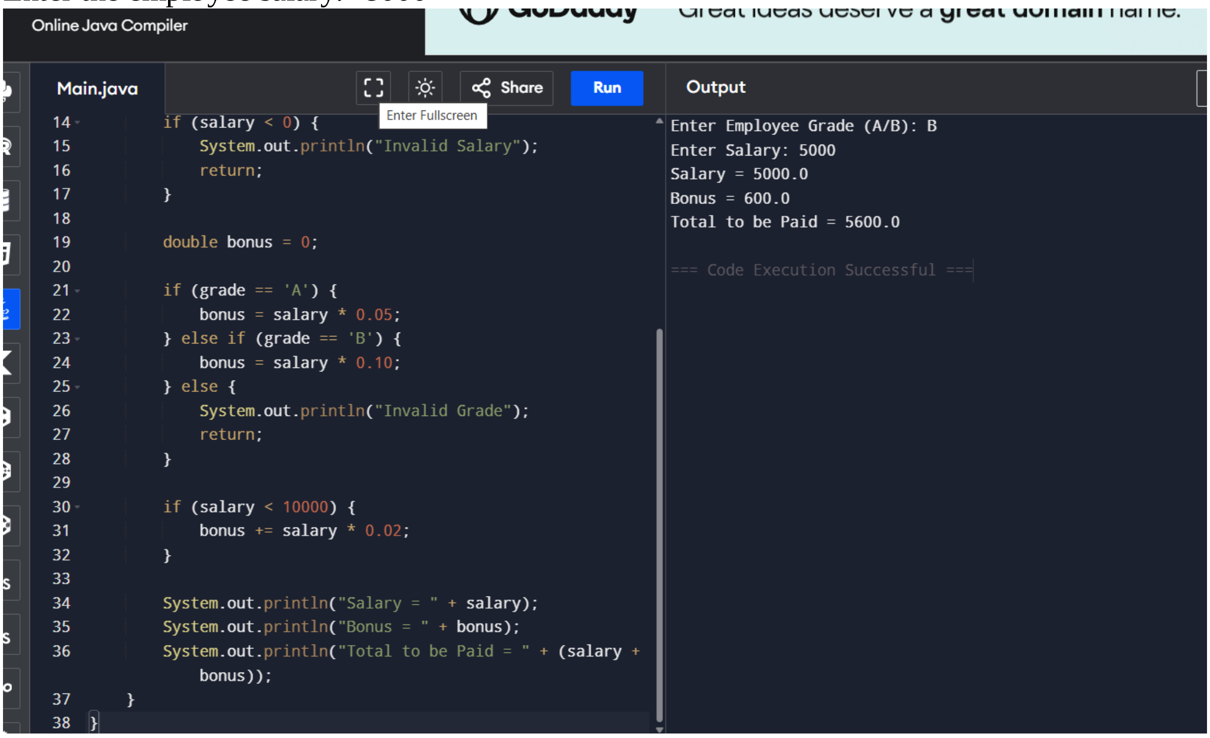
6. In an organization they decide to give bonus to all the employees on New Year. A 5% bonus on salary is given to the grade A workers and 10% bonus on salary to the grade B workers. Write a program to enter the salary and grade of the employee. If the salary of the employee is less than $10,000 then the employee gets an extra 2% bonus on salary Calculate the bonus that has to be given to the employee and print the salary that the employee will get.

   Sample Input & Output:
   Enter the grade of the employee: B
   Enter the employee salary: 50000
   Salary=50000
   Bonus=5000.0
   Total to be paid:55000.0

Test cases:

1. Enter the grade of the employee: A
   Enter the employee salary: 8000
2. Enter the grade of the employee: C
   Enter the employee salary: 60000
3. Enter the grade of the employee: B
   Enter the employee salary: 0
4. Enter the grade of the employee: 38000
   Enter the employee salary: A
5. Enter the grade of the employee: B
   Enter the employee salary: -8000

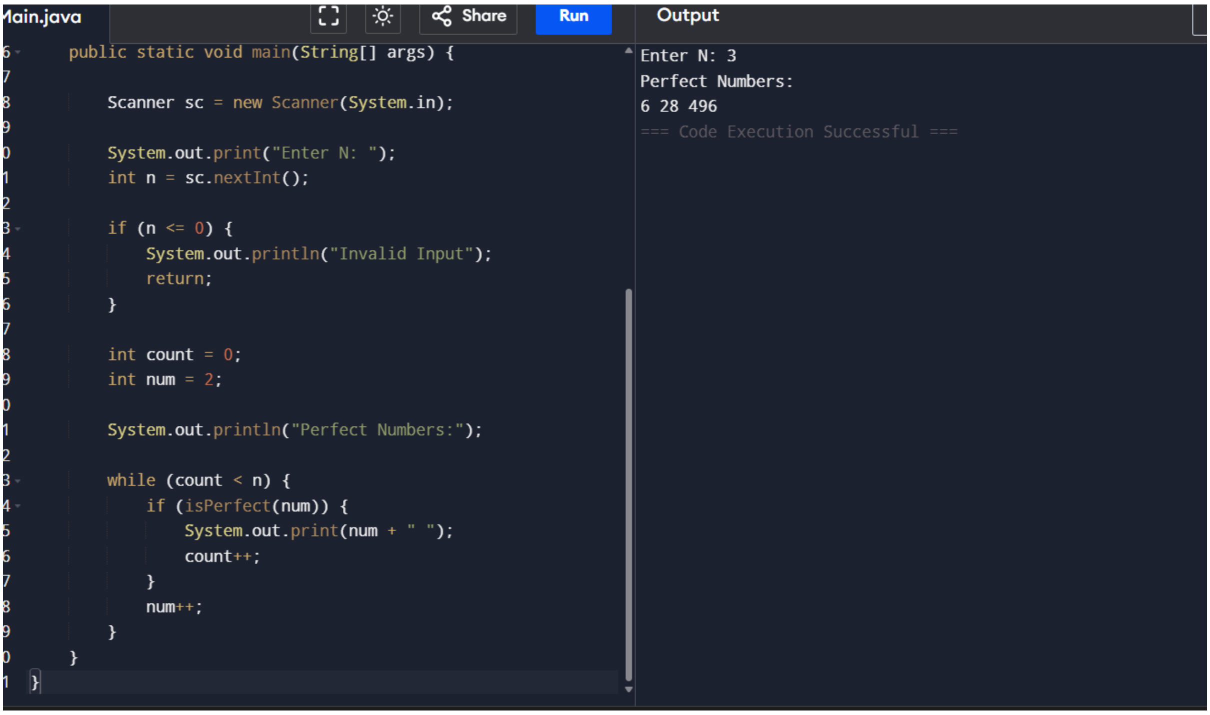



```java
        if (salary < 0) {
            System.out.println("Invalid Salary");
            return;
        }

        double bonus = 0;

        if (grade == 'A') {
            bonus = salary * 0.05;
        } else if (grade == 'B') {
            bonus = salary * 0.10;
        } else {
            System.out.println("Invalid Grade");
            return;
        }

        if (salary < 10000) {
            bonus += salary * 0.02;
        }

        System.out.println("Salary = " + salary);
        System.out.println("Bonus = " + bonus);
        System.out.println("Total to be Paid = " + (salary +
            bonus));
    }
}
```

```
Enter Employee Grade (A/B): B
Enter Salary: 5000
Salary = 5000.0
Bonus = 600.0
Total to be Paid = 5600.0

=== Code Execution Successful ===
```

7. Write a program to print the first n perfect numbers. (Hint Perfect number means **a positive integer that is equal to the sum of its proper divisors)**

Sample Input:
N = 3
Sample Output:
First 3 perfect numbers are: 6 , 28 , 496
Test Cases:
1. N = 0
2. N = 5
3. N = -2
4. N = -5
5. N = 0.2




```java
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter N: ");
        int n = sc.nextInt();

        if (n <= 0) {
            System.out.println("Invalid Input");
            return;
        }

        int count = 0;
        int num = 2;

        System.out.println("Perfect Numbers:");

        while (count < n) {
            if (isPerfect(num)) {
                System.out.print(num + " ");
                count++;
            }
            num++;
        }
    }
}
```

```
Enter N: 3
Perfect Numbers:
6 28 496
=== Code Execution Successful ===
```

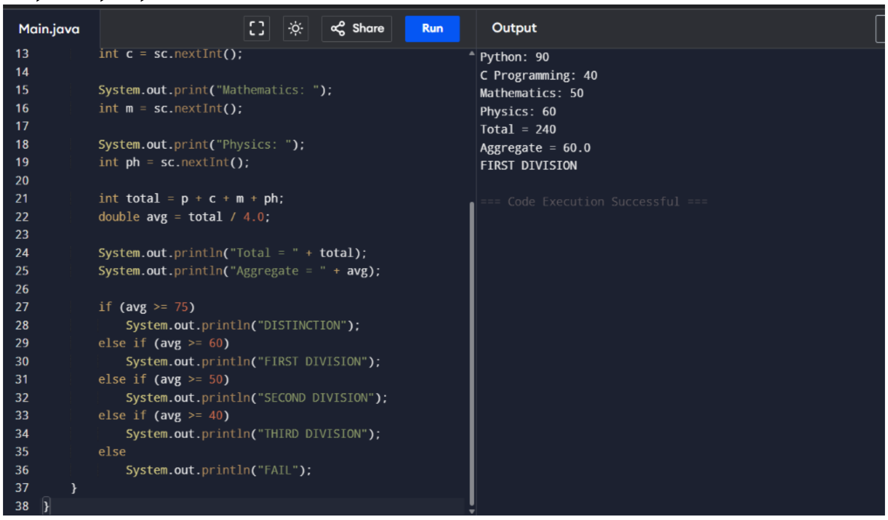
8. Write a program to enter the marks of a student in four subjects. Then calculate the total and aggregate, display the grade obtained by the student. If the student scores an aggregate greater than 75%, then the grade is Distinction. If aggregate is 60>= and <75, then the grade is First Division. If aggregate is 50 >= and <60, then the grade is Second Division. If aggregate is 40>= and <50, then the grade is Third Division. Else the grade is Fail.

    Sample Input & Output:
    Enter the marks in python: 90
    Enter the marks in c programming: 91
    Enter the marks in Mathematics: 92
    Enter the marks in Physics: 93
    Total= 366
    Aggregate = 91.5
    DISTINCTION

    Test cases:

    a) 18, 76,93,65

b) 73,78,79,75
c) 98,106,120,95
d) 96,73, -85,95
e) 78,59.8,76,79



9. Write a program to read the numbers until -1 is encountered. Find the average of positive numbers and negative numbers entered by user.

Sample Input:

Enter -1 to exit…
Enter the number: 7
Enter the number: -2
Enter the number: 9
Enter the number: -8
Enter the number: -6
Enter the number: -4
Enter the number: 10
Enter the number: -1

Sample Output:

The average of negative numbers is: -5.0
The average of positive numbers is : 8.66666667

Test cases:

1. -1,43, -87, -29, 1, -9
2. 73, 7-6,2,10,28,-1
3. -5, -9, -46,2,5,0
4. 9, 11, -5, 6, 0,-1
5. -1,-1,-1,-1,-1

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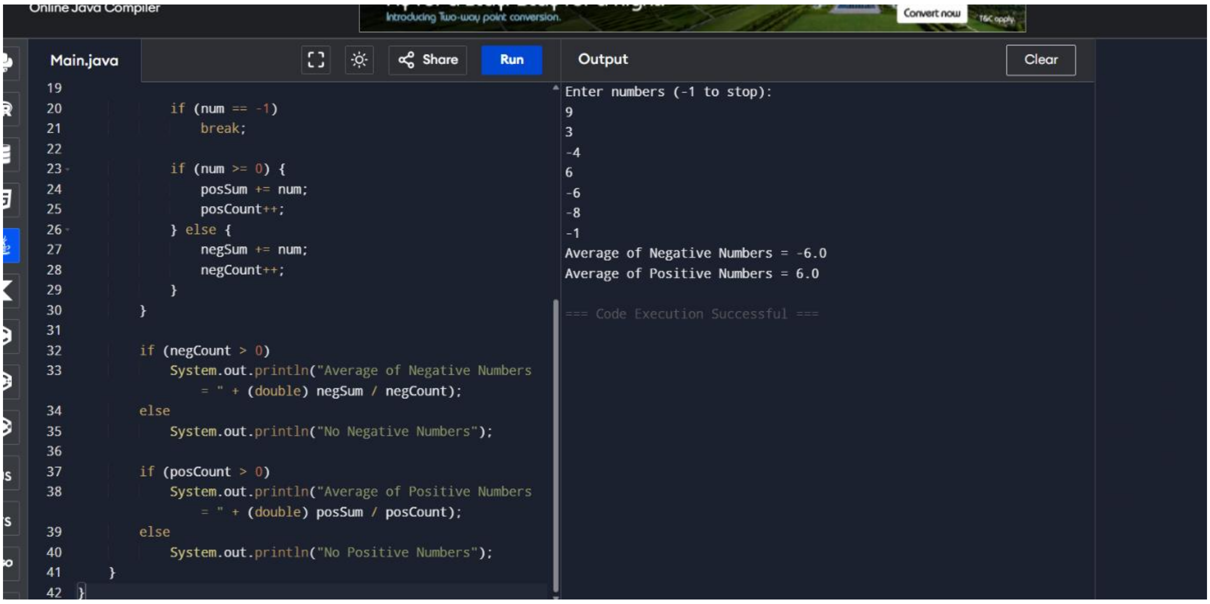

10. Write a program to read a character until a ***** is encountered. Also count the number of uppercase, lowercase, and numbers entered by the users.

    Sample Input:
    Enter * to exit…
    Enter any character: W
    Enter any character: d
    Enter any character: A
    Enter any character: G
    Enter any character: g
    Enter any character: H
    Enter any character: *
    Sample Output:
    Total count of lower case:2
    Total count of upper case:4
    Total count of numbers =0
    Test cases:
    1. 1,7,6,9,5
    2. S, Q, l, K,7, j, M
    3. M, j, L, &, @, G
    4. D, K, I, 6, L, *
    5. *, K, A, e, 1, 8, %, *

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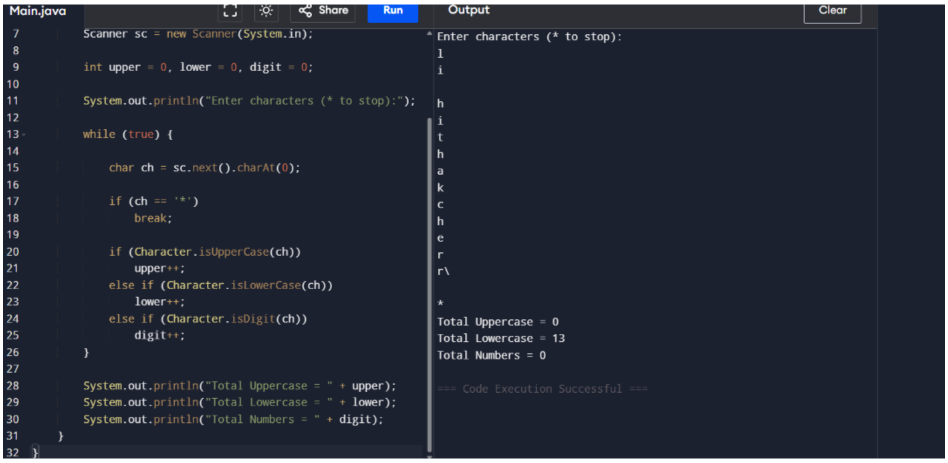


11. Write a program to calculate the factorial of number using recursive function.

Sample Input & Output:

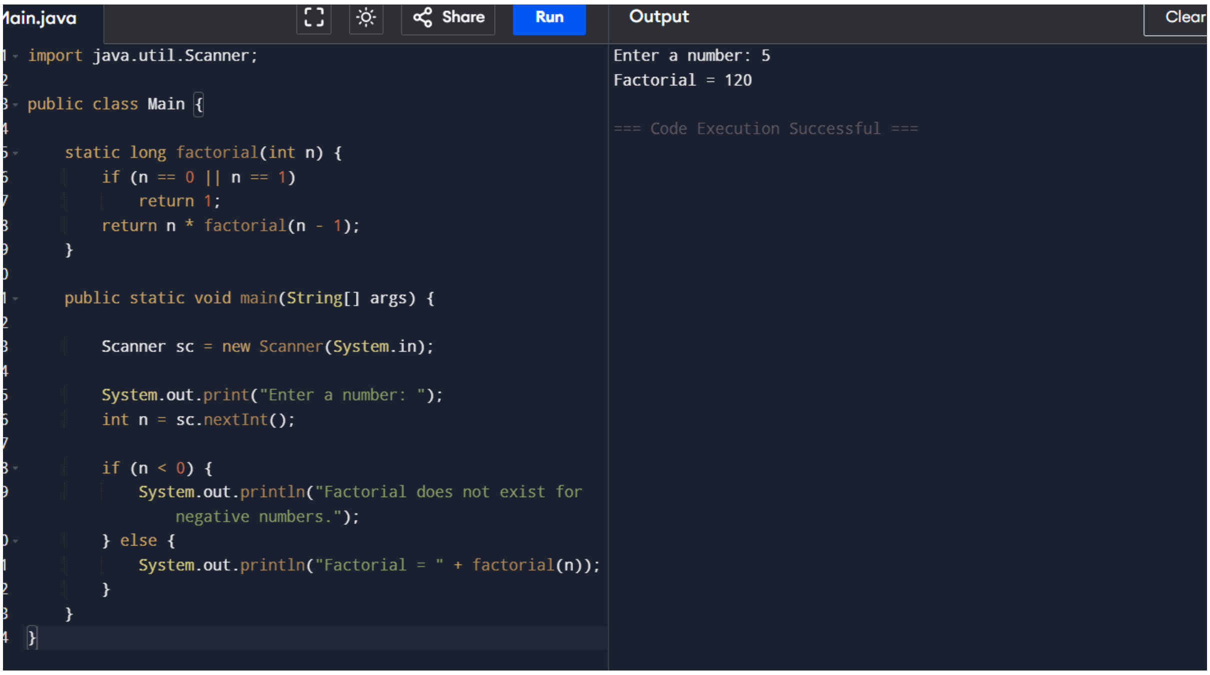
Enter the value of n: 6

Sample Input & Output:

The factorial of 6 is: 720

Test cases:

1. N = 0
2. N = -5
3. N = 1
4. N = M
5. N = %



12. Write a Program to Find the Nth Largest Number in a array.

Sample Input:

List : {14, 67, 48, 23, 5, 62}

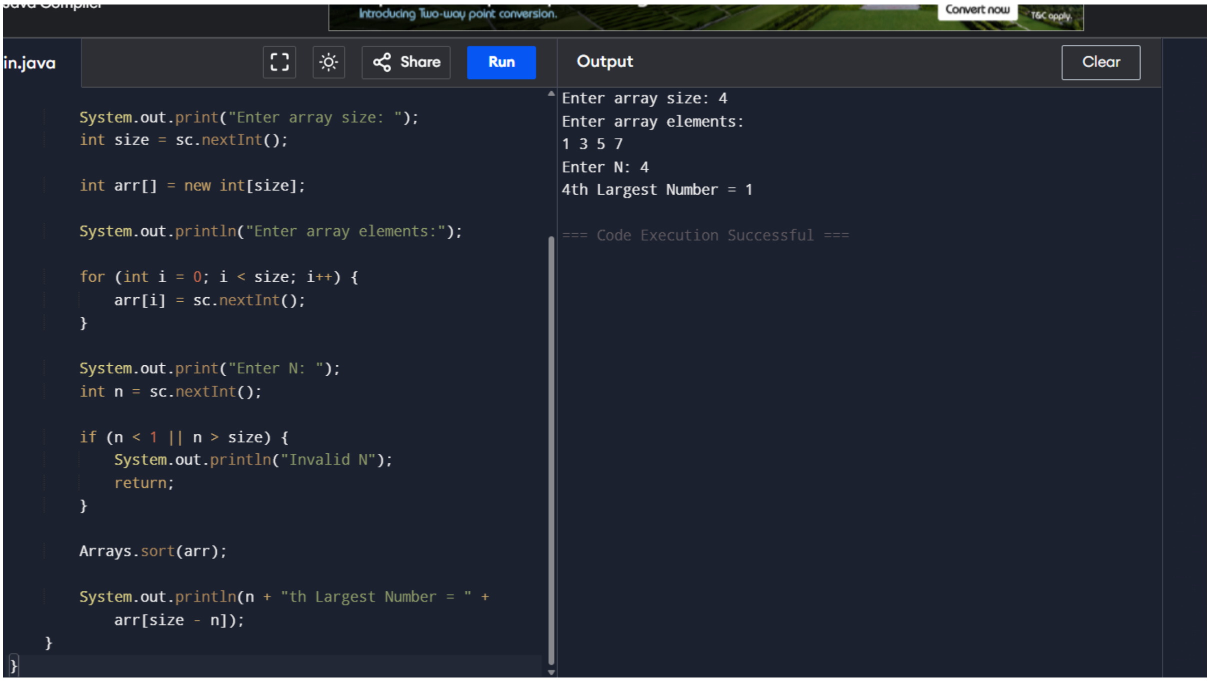
N = 4

Sample Output:

4th Largest number: 23

Test cases:

1. N = 0
2. N = -5
3. N = 1
4. N = M
5. N = %



13. Write a program to convert the Binary to Decimal, Octal

Sample Input:

Given Number: 1101

Sample Output:

Decimal Number: 13

Octal:15

Test cases:

1. 211
2. 11011
3. 22122
4. 111011.011
5. 1010.0101

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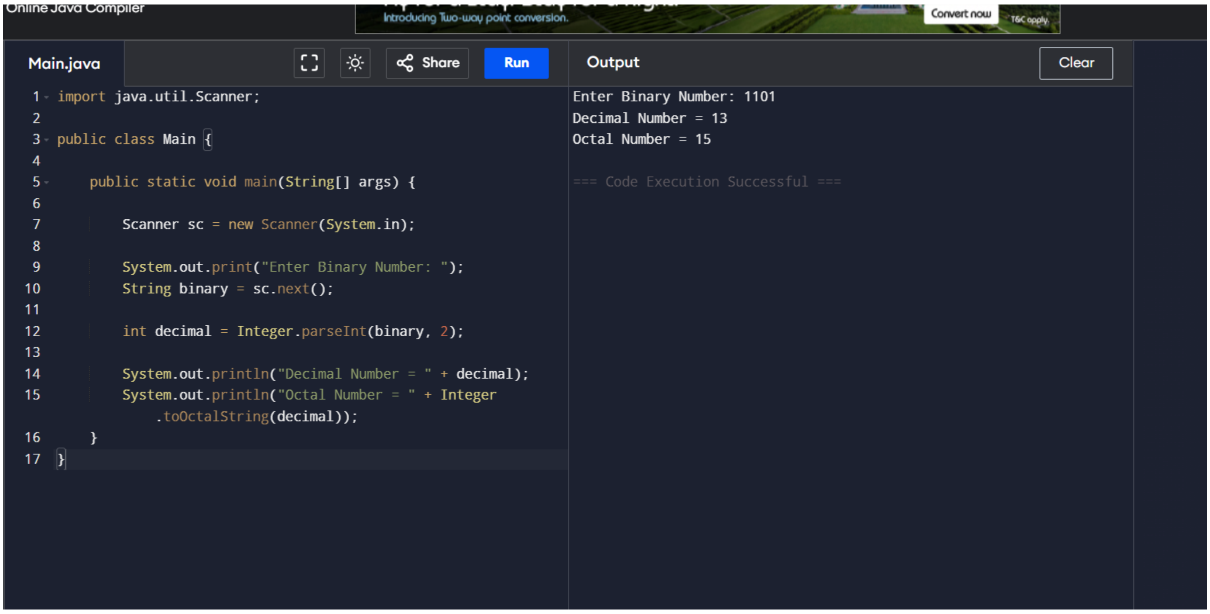
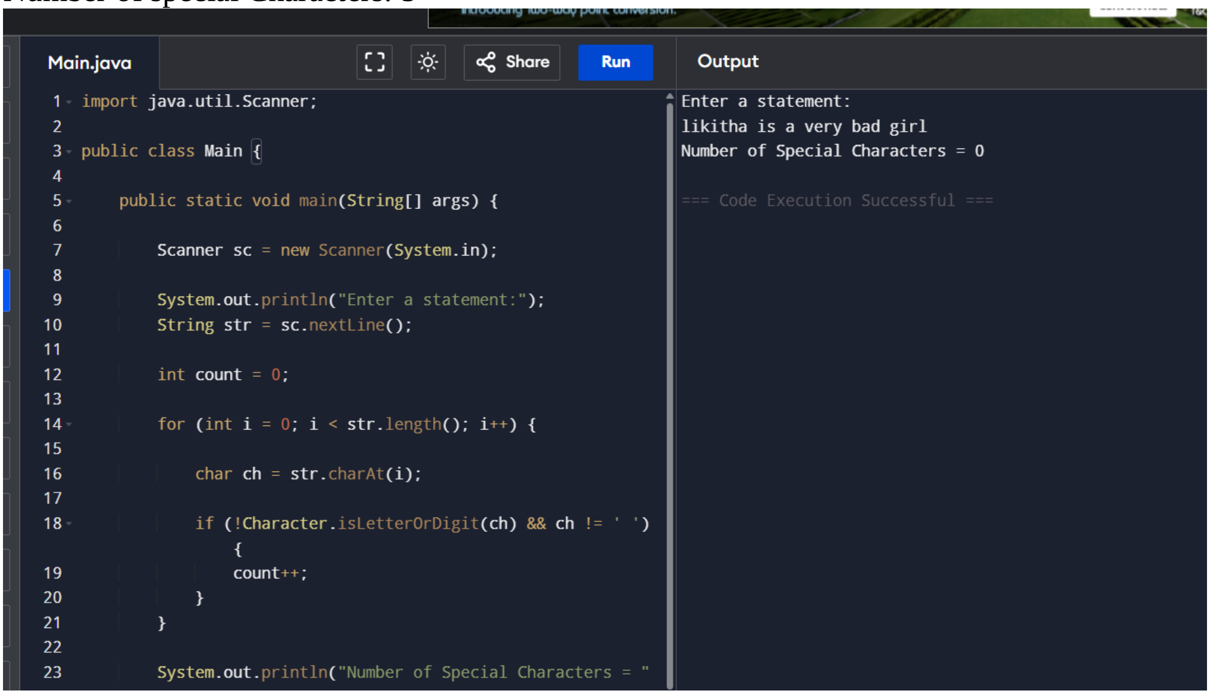


14. Write a program to find the number of special characters in the given statement
    Sample Input:
    Given statement: Modi Birthday @ September 17, #&$% is the wishes code for him.
    Sample Output:
    Number of special Characters: 5



15. Write a Program to Remove the Duplicate Items from a array.
    Sample Input:
        Enter the number of elements in array:7
        Enter element1:10

Enter element2:20
Enter element3:20
Enter element4:30
Enter element5:40
Enter element6:40
Enter element7:50

Sample Output:

Non-duplicate items:
[10, 20, 30, 40, 50]